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Original Article

Robust detection of watermarks for large language models under human edits

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Abstract

Watermarking is an effective approach to distinguishing text generated by large language models (LLMs) from human-written text. However, the pervasive presence of human edits on LLM-generated text dilutes watermark signals, thereby significantly degrading detection performance of existing methods. In this paper, by modelling human edits through mixture model detection, we introduce a new method—a truncated goodness-of-fit test (Tr-GoF) for detecting watermarked text under human edits. We prove that Tr-GoF achieves optimality in robust detection of the Gumbel-max watermark in a certain asymptotic regime of substantial text modifications and vanishing watermark signals. Importantly, Tr-GoF achieves this optimality adaptively without requiring precise knowledge of human edit levels or probabilistic specifications of LLMs, unlike the optimal but impractical Neyman–Pearson likelihood ratio test. Moreover, we establish that Tr-GoF attains the highest detection efficiency rate under moderate text modifications. In contrast, sum-based detection rules used by existing methods fail to achieve optimal robustness in both regimes because the additive nature of their statistics is less resilient to edit-induced noise. We demonstrate Tr-GoF's competitive and sometimes superior performance on synthetic data and open-source LLMs in the OPT and LLaMA families.

Keywords: adaptive optimality, detection efficiency, large language model, phase transition, robust detection, watermark

1 Introduction

Large language models (LLMs) have recently emerged as a transformative technique for generating human-like text and other media (Achiam et al., 2024; OpenAI, 2023; Touvron et al., 2023). While this advancement boosts productivity across various industries, it also introduces risks related to the ownership and creation of content. These risks include the spread of misinformation (Starbird, 2019; Weidinger et al., 2022; Zellers et al., 2019), challenges to education and academic integrity (Milano et al., 2023; Stokel-Walker, 2022), and issues concerning data authenticity (Radford et al., 2023; Shumailov et al., 2023). These problems highlight the urgent need for methodologies to authenticate and verify the origin of text, specifically, determining whether it is generated by LLMs or humans.

Watermarking text during generation by LLMs has offered a principled and viable approach to resolving these issues (Aaronson, 2023; Christ et al., 2024; Kirchenbauer et al., 2023). Since 2023, a variety of watermarking schemes have been introduced (Dathathri et al., 2024; Fernandez et al., 2023; Fu et al., 2024; Giboulot & Teddy, 2024; Hu et al., 2024; Kuditipudi et al., 2024; Y. Liu &

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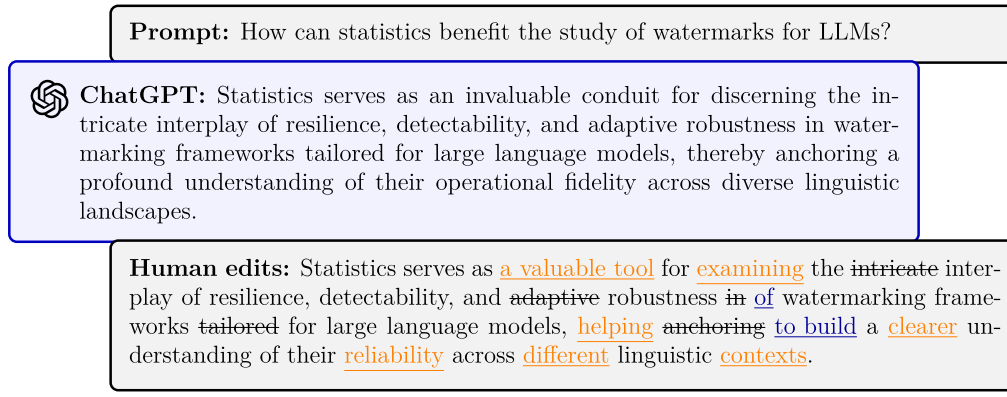


Figure 1. How users modify ChatGPT's response through edits such as substitution, deletion, and insertion. Orange underline marks substitutions, ~~black strikethrough~~ indicates deletions, and blue underline highlights insertions.

Bu, 2024; Wu et al., 2023; Xie et al., 2025; Zhao et al., 2024, 2025). Loosely speaking, a watermarking scheme embeds signals into the process of generating text by using controlled pseudorandomness. This is made possible by the probabilistic nature of LLMs through next-token prediction (NTP) for sequentially generating text. The watermark signals are nearly unnoticeable, if possible, to human readers but are provably detectable once a verifier has knowledge of how the pseudorandomness is constructed (Christ et al., 2024).

In real-world scenarios, however, text generated from LLMs often undergoes various forms of human edits before its use (see Figure 1 for an illustration), which presents perhaps the most significant challenge for applying watermarks (Editorial, 2024; Kirchenbauer et al., 2024). In the simplest case, a user of LLMs might replace several words of the generated text, either to improve the text from the user's perspective or make it less LLM-sounding. This simple editing process would weaken the watermark since the signals from the modified tokens—the smallest unit in text generation, which can be a word or a sub-word—turn to noise. Worse, how the editing process proceeds is unknown to the verifier. For example, it is not clear if a token in the text is generated by the LLM or has been modified by a human. In light of this, a watermark detection rule should not only be efficient in the edit-free scenario but also, perhaps more importantly, its detection power should not be unduly affected by various forms of human edits. Equally important is adaptivity: the rule should perform well without prior knowledge of how or to what extent the text has been edited, ensuring practical applicability in handling diverse and unpredictable editing behaviours.

To get a handle on how robust and adaptive existing watermark detection rules are against human edits, we conduct a numerical experiment and present the results in Figure 2. The experiment is concerned with the Gumbel-max watermark (Aaronson, 2023), which has been implemented internally at OpenAI and is the first unbiased[1] watermark. With 5% of the tokens modified by humans, the detection power when text length is 400 drops from 87.8% in the edit-free case to 64.7% with paraphrase edits and further to 30.2% with adversarial edits. While this significant performance degradation is concerning, it is not surprising as these detection rules (Aaronson, 2023; X. Li et al., 2025) are developed without any robustness consideration, in particular, assuming there is no signal corruption due to human edits. More critically, the editing process would render the computation of pseudorandomness, which takes as input the before-edited tokens (Aaronson, 2023; Kirchenbauer et al., 2023), impossible for detecting watermarks.

Thus, there is a pressing need to develop robust and adaptive methods for detecting watermarks in LLM-generated text that may undergo human edits. Existing work on watermark detection typically assumes a scenario where each token contributes a signal to the watermark. For example,

[1] A watermarking scheme is (approximately) unbiased if the watermarked LLM generates each token with (approximately) the same probability distribution as the unwatermarked counterpart.

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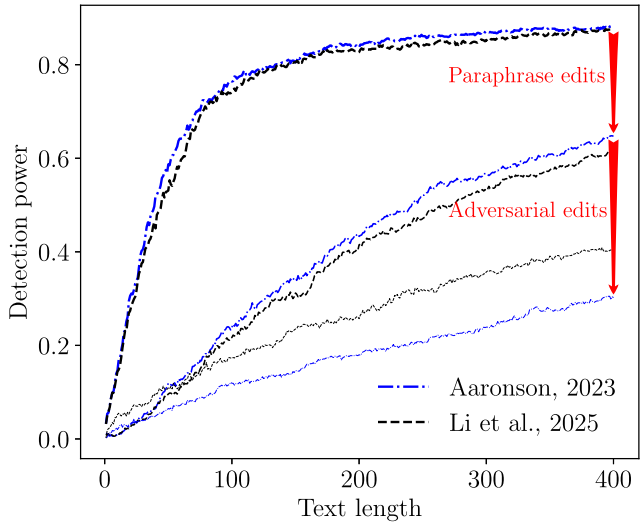


Figure 2. Empirical study of detection methods for the Gumbel-max watermark. Statistical power is evaluated at a 0.01 significance level using 1,000 prompts from the C4 dataset (Raffel et al., 2020) and 400 generated tokens per prompt with the OPT-1.3B model (Zhang et al., 2022) (temperature 0.3). Paraphrase edits randomly replace 5% of tokens with WordNet synonyms (Miller, 1995), while adversarial edits replace the 5% with the strongest watermark signals. Thinner curves indicate higher levels of human edits. See online supplementary material C.1 for details.

X. Li et al. (2025) introduced a statistical framework for evaluating detection efficiency based on sum-based statistics that measure the cumulative watermark strength across all tokens. However, this framework operates under the ideal assumption that tokens are either fully human-written or entirely generated by an LLM. When applied to human-edited text—a combination of human- and LLM-generated tokens—sum-based statistics may be easily corrupted by noise, leading to substantial degradation in efficiency, as illustrated in Figure 2.

1.1 Our contributions

In this paper, we address the robustness of detecting LLM watermarks from a statistical viewpoint, providing both a statistical framework and methodologies with theoretical guarantees alongside empirical validation.

A central challenge in robust watermark detection is understanding how human edits affect detection performance. Although the exact editing process cannot be recovered from the final modified text, a key insight is that—due to the indistinguishability between pseudorandomness and true randomness—each token either contributes a watermark signal or acts as noise, depending on whether it has been altered by human edits. In the framework of X. Li et al. (2025), the noise follows a fixed and known distribution, while the signal depends on the token distribution predicted by the LLM. We refer to this predictive multinomial distribution as the NTP distribution, denoted by $\boldsymbol{P}_t = (P_{t,w})_{w\in\mathcal{W}}$, where $\mathcal{W}$ is the vocabulary used for predicting the tth token. For instance, GPT-2/3.5 models use a vocabulary of size 50,257 (Brown et al., 2020; Radford et al., 2019), while the LLaMA-2 models use 32,000 tokens (Touvron et al., 2023). Motivated by this binary behaviour, we model the watermark signal across all tokens as a mixture: with probability ε_n, a token carries signal; otherwise, it contributes noise. This leads to a natural statistical question: under what conditions can we reliably distinguish this mixture (representing watermarked text potentially modified by humans) from a pure noise distribution (representing entirely human-written text)? We address this question in the following result, illustrated, for example, in the case of the Gumbel-max watermark. (We write $a_n \asymp b_n$ to mean there exist constants C_1 and C_2 such that $C_1 a_n \le b_n \le C_2 a_n$ for all n):

Phase transition for watermark detection. We consider an asymptotic regime where the text length n tends to infinity, but individual tokens contain less watermark signal. Specifically, we

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assume $\varepsilon_n \asymp n^{-p}$ and $1 - \max_{w \in \mathcal{W}} P_{t,w} \asymp n^{-q}$ for constants $0 < p, q < 1$, inspired by the sparse mixture detection problem Donoho and Jin (2004). Detection becomes more challenging as either p or q increases. A larger value of p means fewer tokens contribute to the watermark after human editing, resulting in a diluted watermark; similarly, a larger value of q indicates that the NTP distributions are less regular due to being approximately degenerate, which weakens watermark signals (see elaboration in Section 4.1). Our finding is that the optimal detection region in the (p, q) plane has boundary $q + 2p = 1$. Explicitly, when $q + 2p > 1$, no test can asymptotically achieve zero Type I and Type II errors for the Gumbel-max watermark; when $q + 2p < 1$, in stark contrast, both Type I and Type II errors approach zero using the likelihood-ratio test.

While the likelihood-ratio test achieves theoretical optimality, it requires knowledge of the watermark fraction ε_n and all NTP distributions $\boldsymbol{P}_t$ for $t = 1, \ldots, n$. To make this boundary practically relevant, it is essential to develop a practical method that can reliably separate the null and alternative distributions as $n \to \infty$. For this purpose, it is instructive to examine the statistical limits of sum-based test statistics as employed in X. Li et al. (2025), Kuditipudi et al. (2024) and Fernandez et al. (2023). We prove that these sum-based detection rules fail to achieve the optimal boundary $q + 2p = 1$ in general, and more precisely, the best possible detection boundary these methods can achieve is $q + p = 1/2$.

This analysis offers insights into the causes of this suboptimality, which we leverage to develop a new, practically adaptive method that achieves the optimal boundary:

An adaptively optimal method. Our approach utilizes the empirical cumulative distribution function of p-values from the text under detection, comparing it with the null counterpart through a form of divergence (Jager & Wellner, 2007). Unlike sum-based approaches X. Li et al. (2025), Kuditipudi et al. (2024) and Fernandez et al. (2023), this goodness-of-fit test adaptively identifies areas of significant departure between null and alternative distributions without requiring prior knowledge of the watermark fraction or NTP regularity. Moreover, a crucial innovation is to truncate the search for the location to filter out extreme scores for stability. This gives a family of truncated goodness-of-fit tests, which we refer to as Tr-GoF for short (details in Algorithm 1). We show that Tr-GoF achieves the robust detection boundary $q + 2p = 1$ without relying on any tuning parameters, thereby possessing adaptive optimality Donoho and Jin (2004, 2015). Figure 3 shows the robustness of Tr-GoF in detecting the Gumbel-max watermark in human-edited text.

Algorithm 1 Truncated GoF detection method (Tr-GoF)

1: **Input:** Edited text $w_{1:n}$, hash function $\mathcal{A}$, secret key $\texttt{Key}$, pivot statistic function Y, stability parameter c_n^+, and critical value δ.

2: For $t = 1, 2, \ldots, n$, compute the pseudorandomness number $\zeta_t = \mathcal{A}(w_{(t-m):(t-1)}, \texttt{Key})$.

3: For $t = 1, 2, \ldots, n$, compute the pivot statistic $Y_t = Y(w_t, \zeta_t)$.

4: For $t = 1, 2, \ldots, n$, calculate the p-value as $p_t = 1 - Y_t$.

5: Sort the p-values in the ascending order $p_{(1)} < p_{(2)} < \ldots < p_{(n)}$ and set $p_{(n+1)} = 1$.

6: Define the function K_s^+ as in (6) and compute the test statistic by

$$S_n^+(s) = \sup_{t:\, p_{(t+1)} \geq c_n^+} K_s^+(t/n,\ p_{(t)}).$$

7: **Claim:** Text $w_{1:n}$ is partially LLM-generated if (8) holds; otherwise, it is human-written.

Having established the optimality of Tr-GoF for sparse watermark detection, we further examine its performance in a regime of 'dense' watermark signals. More precisely, we consider the setting where the watermark fraction $\varepsilon_n \equiv \varepsilon$ remains constant, and the regularity level of the NTP distribution satisfies $1 - \max_{w \in \mathcal{W}} P_{t,w} \geq \Delta$ for some constant $0 < \Delta < 1$. Following is our third main result in this paper:

Tr-GoF attains optimal efficiency rate. We show that the Tr-GoF test achieves the highest class-dependent efficiency under the framework of watermarks for LLMs proposed by X. Li et al.

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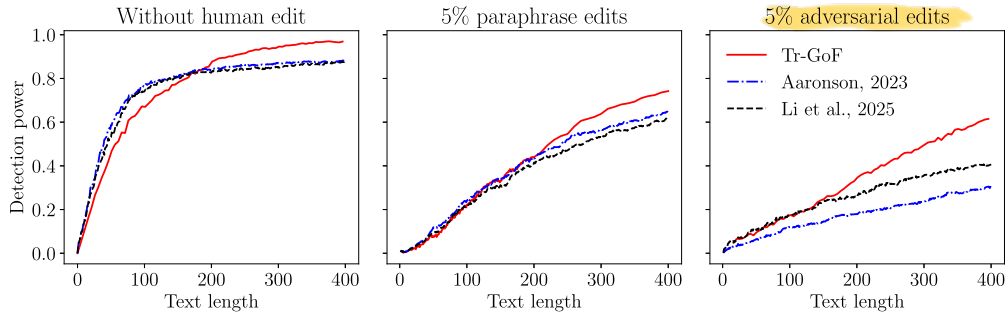


Figure 3. Tr-GoF shows improved detection efficiency under the same setup as Figure 2, with a more pronounced performance advantage in adversarial edits compared to paraphrase edits.

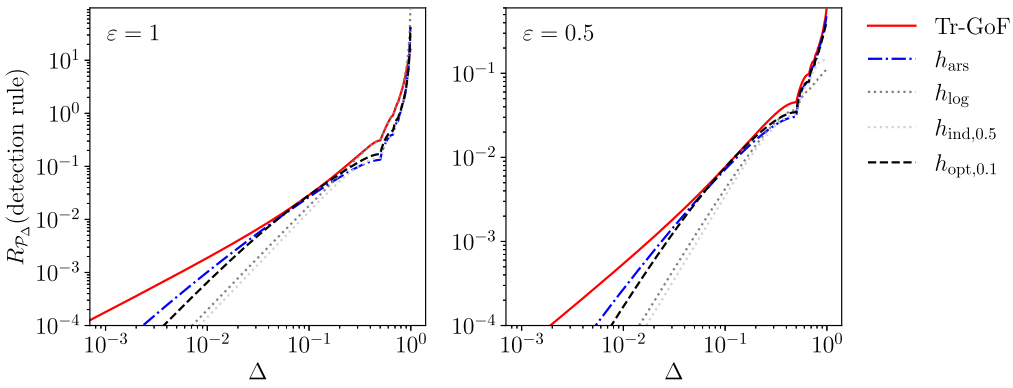


Figure 4. $\mathcal{P}_\Delta$-efficiency rates of different detection methods for the Gumbel-max watermarks. $R_{\mathcal{P}_\Delta}$ has non-smooth points when $\Delta = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$ (X. Li et al., 2025).

(2025). This optimality is adaptive in that it does not rely on any prior knowledge of Δ or ε, whereas existing sum-based detection rules fail to achieve the same level of efficiency due to their lack of adaptivity for dense watermark detection (see Figure 4 for illustration). This finding is unexpected as the Tr-GoF test was originally motivated by sparse watermark detection. Taken together, our results show that the Tr-GoF test achieves optimality in both sparse and dense regimes of watermark detection.

1.2 Related work

Most robust watermarking methods focus on algorithmic or cryptographic approaches. Algorithmic methods emphasize resilient watermarking designs (Giboulot & Teddy, 2024; Hou et al., 2024; Kuditipudi et al., 2024; Ren et al., 2024; Yoo et al., 2023; Zhu et al., 2024), while cryptographic approaches protect watermark signals through robust pseudorandom codes (Christ & Gunn, 2024; Golowich & Moitra, 2024). However, rigorous statistical analysis of robustness in watermark detection remains limited, largely due to the absence of a cohesive statistical framework. Definitions of robustness also vary widely.

Our approach introduces a statistical perspective by modelling human edits as a mixture distribution, enabling the establishment of detection boundaries and class-dependent efficiency rates. Another statistical approach to robustness is the edit tolerance limit, which quantifies the maximum proportion of edits a detection method can withstand while remaining effective. Studies often apply this perspective to the green-red list watermark, initially proposed in (Kirchenbauer et al., 2023, 2024) and valued for its simplicity (Fernandez et al., 2023; Hu et al., 2024; A. Liu et al., 2024; Wu et al., 2023; Zhu et al., 2024). Zhao et al. (2024) further improved its tolerance

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by introducing a fixed pseudorandom code, enhancing resilience against text modifications. Additional refinements of this approach are discussed in Kuditipudi et al. (2024), Cai et al. (2024) and Huang et al. (2023).

Other forms of robustness include decoder-focussed and cryptographic approaches. Zhao et al. (2025) examines decoder functions that map logit vectors to token probabilities, proposing a provably robust decoder for enhanced resilience. In cryptographic studies, robustness is designed to make pseudorandom codes resistant to a fixed proportion of human edits (Christ & Gunn, 2024; Golowich & Moitra, 2024). Our statistical viewpoint complements these approaches, providing an orthogonal framework for assessing robustness.

Robust statistics, which traditionally addresses resilience to outliers and perturbations, provides a relevant foundation for our study (Diakonikolas & Kane, 2023; Huber, 1992; Huber & Ronchetti, 2011). Our mixture formulation connects with Huber's contamination model (Huber, 1992); however, unlike in Huber's model where the contaminated distribution is typically unknown, in watermark detection, this distribution is known. Additionally, our work is related to sparse mixture detection, first introduced in (Dobrusin, 1958) to identify sparse signals, with further developments in mixtures model (Arias-Castro & Wang, 2017; Cai et al., 2011; Cai & Wu, 2014; Donoho & Jin, 2004; Ingster, 1996) and rare-signal models (Jin & Ke, 2016; Jin et al., 2021; Ke et al., 2014). Robust watermark detection also diverges from traditional sparse detection due to the autoregressive nature of text generation, which results in time-varying signal (or NTP) distributions. While some methods address inhomogeneity using stationary Gaussian processes (Hall & Jin, 2008) or temporal correlations (Hall & Jin, 2010), they are inapplicable to text generation. Our theoretical analysis directly leverages the autoregressive nature to analyse inhomogeneity in watermark detection, as detailed in Section 4.

1.3 Organization of the paper

The remainder of the paper is organized as follows. In Section 2, we introduce basics of LLMs and the statistical framework of LLM watermarks (X. Li et al., 2025). In Section 3, we develop the Tr-GoF test and establish its theoretical guarantees for robust detection in Section 4. We evaluate the empirical performance of the Tr-GoF test in Sections 5 and 6. We conclude in Section 7 with a discussion of future research directions. Most technical proofs are provided in the supplementary. Source code is publicly available at https://github.com/lx10077/TrGoF.

2 Preliminaries

Watermarking protocol. We consider a standard watermarking protocol involving three parties (Kuditipudi et al., 2024; Xie et al., 2025): the model provider, the user, and the verifier. For example, a student (the user) may use an LLM to assist with a homework assignment. The model provider embeds a statistical watermark into the generated text using a secret key `Key`, which is shared with the verifier (e.g. a teacher), but not with the user. The user may modify the generated text before submitting it. The verifier, who observes only the final (possibly edited) text, attempts to determine whether it contains a watermark. Importantly, the verifier does not have access to the original prompt, the unedited generated text, or the internal NTP distributions used by the model. This protocol unfolds over three stages: first, watermark embedding, where the model encodes a hidden signal into the generated text; second, human editing, where the user may modify the output; and finally, watermark detection, where the verifier analyses the submitted text to detect any remaining watermark signal.

Watermark embedding. We begin by describing the embedding mechanism. LLMs, such as GPT models (Brown et al., 2020; Radford et al., 2019), generate text by sampling tokens autoregressively. At each position t, given the previously generated tokens $w_{1:(t-1)} := w_1 w_2 \cdots w_{t-1}$, the model computes a next-token prediction (NTP) distribution $\boldsymbol{P}_t = (P_{t,w})_{w \in \mathcal{W}}$ and samples the next token w_t accordingly. To embed a watermark, this sampling process is modified in a deterministic yet secret-key-dependent way. Specifically, at each step t, the model computes a pseudorandom variable $\zeta_t = \mathcal{A}(w_{(t-m):(t-1)}, \texttt{Key})$, where $\mathcal{A}$ is a cryptographic hash function, `Key` is the watermarking key, and m is the context window size. The token is then deterministically selected as $w_t = \mathcal{S}(\boldsymbol{P}_t, \zeta_t)$ using a decoding function $\mathcal{S}$. The variable ζ_t is termed pseudorandom because it behaves like a

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random variable, yet can be deterministically reconstructed by the verifier from the observed tokens $w_{1:n}$, the key Key, and the hash function $\mathcal{A}$. For theoretical analysis, we model ζ_t as i.i.d. samples from a reference distribution π, reflecting the cryptographic properties of $\mathcal{A}$ (Barak, 2021; Schneier, 1996; Wu et al., 2023; Zhao et al., 2025). A decoder $\mathcal{S}$ is said to be unbiased if it preserves the original distribution in expectation, i.e. $\mathbb{P}_{\zeta\sim\pi}(\mathcal{S}(\boldsymbol{P}, \zeta) = w) = P_w$ for any $\boldsymbol{P}$ and token w. By this definition, an unbiased decoder corresponds to standard multinomial sampling methods.

Watermark detection. Since our primary interest lies in designing and analysing detection methods, we adopt the perspective of the verifier. With a slight abuse of notation, we refer to the observed (potentially edited) text still as $w_{1:n}$. During watermark detection, the verifier receives the text $w_{1:n}$ and reconstructs the corresponding pseudorandom sequence $\zeta_{1:n} := \zeta_1\zeta_2\cdots\zeta_n$ using the shared key and hash function. If the text is entirely human-written, there should be no statistical dependence between $w_{1:n}$ and $\zeta_{1:n}$, as human users do not know the key or hash function. In contrast, if the text is fully generated by a watermarked LLM, each token w_t is deterministically chosen based on ζ_t through the decoder function $\mathcal{S}$. This observation reduces watermark detection to a problem of testing for statistical dependence between the observed tokens and the pseudorandom variables. X. Li et al. (2025) formalized this idea as a hypothesis testing framework and analysed the effectiveness of various detection methods. The central tool in this framework is a pivotal statistic $Y_t = Y(w_t, \zeta_t)$, where the function $Y(\cdot, \cdot)$ is designed so that the distribution of Y_t is known whenever w_t and ζ_t are independent—regardless of the marginal distribution of w_t. In this way, under the null hypothesis H_0, where the text is entirely human-written, the statistics Y_t are i.i.d. samples from a known reference distribution μ_0. Under the alternative hypothesis H_1, where the text is fully generated by a watermarked LLM, each $Y_t = Y(\mathcal{S}(\boldsymbol{P}_t, \zeta_t), \zeta_t) \mid \boldsymbol{P}_t$ follows a distribution that depends on the NTP distribution $\boldsymbol{P}_t$, denoted by $\mu_{1,\boldsymbol{P}_t}$. Watermark detection is thus formulated as the following hypothesis testing problem:

$$H_0 : Y_t \sim \mu_0 \text{ i.i.d. for } 1 \le t \le n \quad \text{v.s.} \quad H_1 : Y_t \mid \boldsymbol{P}_t \sim \mu_{1,\boldsymbol{P}_t} \text{ for } 1 \le t \le n. \tag{1}$$

The Gumbel-max watermark: Decoder and detection. Our paper focuses on the Gumbel-max watermark, one of the most influential and the first unbiased watermark (Aaronson, 2023), which has been implemented internally at OpenAI (OpenAI, 2024) and serves as a baseline in many studies. This watermark utilizes the Gumbel-max trick, a sampling technique for multinomial distributions (Gumbel, 1948; Jang et al., 2016; Maddison et al., 2014; Papandreou & Yuille, 2011). The trick ensures that $\arg\max_{w\in\mathcal{W}} P_w^{-1} \log U_w$ follows the distribution $\boldsymbol{P} = (P_w)_{w\in\mathcal{W}}$, where $\zeta = (U_w)_{w\in\mathcal{W}}$ consists of $|\mathcal{W}|$ i.i.d. $U(0, 1)$ random variables. Scott Aaronson proposed the following unbiased decoder (Aaronson, 2023):

$$\mathcal{S}^{\text{gum}}(\boldsymbol{P}, \zeta) := \arg\max_{w\in\mathcal{W}} \frac{\log U_w}{P_w}. \tag{2}$$

The pivotal statistic is $Y_t = Y(w_t, \zeta_t) = U_{t,w_t}$, where $\zeta_t = (U_{t,w})_{w\in\mathcal{W}}$ collects all pseudorandom numbers. The watermark is detected when the sum-based statistic $T_n^{\text{ars}} = \sum_{t=1}^n h_{\text{ars}}(Y_t)$, with $h_{\text{ars}}(y) = -\log(1-y)$, exceeds a set threshold. The method works because, without a watermark, the U_{t,w_t}'s are i.i.d. $U(0, 1)$, so $Y_t \sim \mu_0 = U(0, 1)$. In contrast, if the watermark is present, (2) makes tokens with a larger pseudorandom number more likely to be selected. Indeed, we have $Y_t \mid \boldsymbol{P}_t \sim \mu_{1,\boldsymbol{P}_t}$, where $\mu_{1,\boldsymbol{P}}(Y \le r) = \sum_{w\in\mathcal{W}} P_w r^{1/P_w}$ for $r \in [0, 1]$ (X. Li et al., 2025). This alternative distribution is distinct from μ_0 unless $\boldsymbol{P}$ is degenerate (i.e. concentrated on a single token). Detection thus hinges on distinguishing the statistical gap between μ_0 and $\mu_{1,\boldsymbol{P}}$. Beyond h_{ars}, other scoring functions have been proposed. Examples include the log function $h_{\log}(y) = \log y$

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(Fernandez et al., 2023; Kuditipudi et al., 2024), and the optimal least-favourable score $h_{\text{opt},\Delta}$ from (X. Li et al., 2025) where Δ is a user-specified regularity parameter. The function $h_{\text{opt},\Delta}$ is optimal under the assumption that each $\boldsymbol{P}_t$ belongs to a Δ-regular class, denoted $\mathcal{P}_\Delta$, defined as:

$$\mathcal{P}_\Delta = \left\{ \boldsymbol{P} : \max_{w \in \mathcal{W}} P_w \leq 1 - \Delta \right\}. \tag{3}$$

X. Li et al. (2025) shows that $h_{\text{opt},\Delta}$ achieves the fastest exponential decay of Type II error at a fixed significance level α when testing against the least favourable distribution in $\mathcal{P}_\Delta$ (see Definition 1).

3 Method

3.1 Problem formulation

In this section, we formally define the robust detection problem and then introduce our proposed test, Tr-GoF, in Section 3.2. As outlined in Section 2, the watermarking process proceeds in three stages: watermark embedding, human editing, and detection. The prior framework of X. Li et al. (2025) assumes no human edits and formulates detection as a binary hypothesis test (1), so it assumes that the entire text $w_{1:n}$ is either fully human-written or generated by a watermarked LLM. However, this formulation does not capture the reality where users edit watermarked content.

To understand the effect of human edits, consider the pivotal statistic $Y_t := Y(w_t, \zeta_t)$, where both the token w_t and the pseudorandom variable ζ_t are derived from the observed (possibly edited) text. Suppose the subtext $w_{(t-m):t}$ is copied exactly from the original fully watermarked text. In this case, both w_t and ζ_t remain unchanged, as $\zeta_t = \mathcal{A}(w_{(t-m):(t-1)}, \texttt{Key})$ is purely determined by its preceding segment $w_{(t-m):(t-1)}$. Therefore, Y_t follows the alternative distribution $\mu_{1,\boldsymbol{P}_t}$, which reflects the presence of a watermark signal. However, if this $(m+1)$-length subtext has been edited—meaning it does not match any corresponding window in the original watermarked text—then either the recomputed ζ_t or the observed w_t differs from the one originally used. Because any cryptographic hash function $\mathcal{A}$ is highly sensitive to input changes, even a minor modification results in a statistically independent output. In this case, ζ_t is no longer associated with the generation of w_t, and the pair (w_t, ζ_t) is independent. By the definition of a pivotal statistic, this implies that $Y_t \sim \mu_0$, the null distribution. These observations motivate a more flexible detection framework. Depending on the fraction of watermark signal—that is, the dependence between w_t and ζ_t—the distribution of Y_t should be a mixture of μ_0 and $\mu_{1,\boldsymbol{P}_t}$. We therefore model the observed Y_t as arising from a mixture distribution:

$$H_0 : Y_t \sim \mu_0, \ \text{i.i.d.} \quad \forall 1 \leq t \leq n \ \text{v.s.} \ H_1^{\text{mix}} : Y_t \mid \boldsymbol{P}_t \sim (1-\varepsilon_n)\mu_0 + \varepsilon_n \mu_{1,\boldsymbol{P}_t}, \quad \forall\, 1 \leq t \leq n. \tag{4}$$

This $H_1^{(\text{mix})}$ implies that, on average, an ε_n-fraction of watermark signals remain intact and thus detectable. When $\varepsilon_n \equiv 1$, the mixture model in (4) reduces to the original (1).

3.2 The Tr-GoF detection method

In this section, we introduce our detection method, Tr-GoF (Algorithm 1), which we apply to the robust watermark detection problem (4). To begin, we explain the rationale behind this approach. The null hypothesis H_0 in (4) assumes that all pivotal statistics $Y_{1:n}$ are drawn from the same distribution μ_0. This indicates that the testing problem in (4) essentially assesses how well the entire data set $Y_{1:n}$ conforms to the null distribution μ_0. Roughly speaking, Tr-GoF rejects the null hypothesis H_0 when the deviation between the empirical distribution of $Y_{1:n}$ and the null distribution μ_0 is sufficiently large.

We now introduce the deviation measure which is based on a specific ϕ-divergence (Jager & Wellner, 2007). Without loss of generality, we assume the data used in Tr-GoF are all p-values, which is always feasible by transforming observations using the CDF of μ_0. In the Gumbel-max watermark, the p-values are linear transformations of $Y_{1:n}$. Given that $\mu_0 = U(0, 1)$, the p-value

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for each observation is:

$$p_t := \mathbb{P}_0(Y \geq Y_t \mid Y_t) = 1 - Y_t.$$

Let $\mathbb{F}_n(r)$ denote the empirical distribution of the p-values,

$$\mathbb{F}_n(r) = \frac{1}{n}\sum_{t=1}^{n} \mathbf{1}_{p_t \leq r} \quad \text{for } r \in [0, 1],$$

and its expected distribution under H_0 is always the uniform $U(0, 1)$, with CDF $\mathbb{F}_0(r) = r$ for $r \in [0, 1]$. Tr-GoF utilizes the following test statistic to measure the deviation between the empirical CDF $\mathbb{F}_n$ and the expected CDF $\mathbb{F}_0$:

$$S_n^+(s) = \sup_{r \in [p^+, 1)} K_s^+(\mathbb{F}_n(r), \mathbb{F}_0(r)). \tag{5}$$

In the above, there are two truncations: the first is on the r-domain, where $p^+ = \sup\{p_{(t)} : p_{(t)} \leq c_n^+\}$ with $c_n^+ \in [0, 1]$ introduced for stability and $p_{(t)}$ is the tth smallest p-values among $Y_{1:n}$. The second is to truncate K_s to K_s^+, which is defined as a truncated version of K_s defined by

$$K_s^+(u, v) = \begin{cases} K_s(u, v), & \text{if } 0 < v < u < 1, \\ 0, & \text{otherwise,} \end{cases} \tag{6}$$

where $K_s(u, v)$ represents the ϕ_s-divergence between Ber(u) and Ber(v):[2]

$$K_s(u, v) = D_{\phi_s}(\text{Ber}(u) \| \text{Ber}(v)) = v\phi_s\left(\frac{u}{v}\right) + (1 - v)\phi_s\left(\frac{1 - u}{1 - v}\right)$$

Here the scalar function $\phi_s(x)$, indexed by $s \in \mathbb{R}$, is convex in x and is defined by Jager and Wellner (2007):

$$\phi_s(x) = \begin{cases} x \log x - x + 1, & \text{if } s = 1, \\ \frac{1 - s + sx - x^s}{s(1-s)}, & \text{if } s \neq 0, 1, \\ -\log x + x - 1, & \text{if } s = 0. \end{cases} \tag{7}$$

In this family, ϕ_s provides a range of examples depending on the value of s. For $s = 1$, $\phi_1(x) = x \log x - x + 1$, which reduces K_1 to the KL divergence: $K_1(u, v) = v \log \frac{u}{v} + (1 - u) \log \frac{1-u}{1-v}$. When $s = 2$, $\phi_2(x) = \frac{1}{2}(x^2 - x - 1)$, yielding $K_2(u, v) = \frac{(u-v)^2}{2v(1-v)}$, a form associated with Higher Criticism (Donoho & Jin, 2004, 2015), as discussed in Remark 4. For $s \neq 0, 1$, we have $K_s(u, v) = \frac{1}{s(1-s)}[1 - u^s v^{1-s} - (1 - u)^s (1 - v)^{1-s}]$. This definition of ϕ_s ensures continuity in s for all $x \in (0, 1)$.

Tr-GoF is based on the test statistic $S_n^+(s)$, which is introduced in (5) and can be straightforwardly computed using (8). This statistic is expected to be small under the null hypothesis H_0 and large under the alternative hypothesis $H_1^{(\text{mix})}$. We define the detection method as follows: for a small $\delta > 0$, we reject H_0 in favour of $H_1^{(\text{mix})}$ if

$$n \cdot S_n^+(s) \geq (1 + \delta) \log \log n. \tag{8}$$

In other words, if our measure $S_n^+(s)$ of deviance from $\mathbb{F}_n$ to $\mathbb{F}_0$ is sufficiently large—satisfying the condition in (8)—then the detection method will conclude that the observed text was (partially) generated by a watermarked LLM rather than being human-written.

[2] Ber(u) denotes a Bernoulli distribution with parameter (or head probability) u.

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Remark 1 (Selection of the critical value). We discuss the selection of the critical value $(1+\delta)\log\log n$. Given the setup in (4), where $Y_{1:n}$ are i.i.d. samples from μ_0, the Type I error can be controlled by carefully choosing this critical value. Notably, this control is independent of any human edits. In our experiments, we determine the critical value using Monte Carlo simulations, a method that is both computationally efficient and effective in controlling the Type I error even with finite sample sizes.

Remark 2 (Differences from previous work). Our Tr-GoF test departs from the GoF test in Jager and Wellner (2007) by using two truncations. First, we truncate K_s to K_s^+ to facilitate theoretical analysis, leveraging the property that $\lim_{n\to\infty}\mathbb{P}_0(p_{(t)} \le t/n+\delta, \forall t\in[n]) = 1$ for any $\delta>0$. Second, we restrict the domain of r to $[p^+, 1)$ instead of $(0, 1)$ to exclude small p-values, such as $p_{(1)}$ and $p_{(2)}$, in defining $S_n^+(s)$. In summary, Jager and Wellner (2007) focus on the untruncated statistic $S_n(s) := \sup_{r\in(0,1)} K_s(\mathbb{F}_n(r), \mathbb{F}_0(r))$ and its theoretical properties under i.i.d. data. In contrast, we propose a truncated $S_n^+(s)$ in settings (watermark detection) where the data is not i.i.d.

Remark 3 (Drawbacks of an untruncated r-domain). As introduced in Remark 2, the untruncated $S_n(s)$ has two main drawbacks. First, its weak convergence is slow: (Jager & Wellner, 2007) show that $n\cdot S_n(s)$ converges weakly to a random variable under H_0 (see their Theorem 3.1), but Gontscharuk et al. (2015) indicate that this convergence rate is extremely slow, suggesting r-domain truncation as a remedy. This observation also motivates our use of Monte Carlo simulations for determining critical values, despite the exact computation available in J. Li and Siegmund (2015) and Moscovich (2023). Second, without removing extreme values from small p-values like $p_{(1)}$ or $p_{(2)}$, $S_n(s)$ is prone to a heavy-tail issue. In fact, it holds that $\mathbb{P}_0(n\cdot S_n(s)\ge z)\ge \frac{1}{2z}$ for large $z>0$ over certain s values. Test statistics with heavy tails are generally undesirable, as they reduce power at stringent significance levels in small samples. Removing these small p-values mitigates the heavy-tail effect significantly in numerical experiments. See online supplementary material A.4 for more detailed discussion.

Remark 4 (Relationship to Higher Criticism). The celebrated Higher Criticism (HC) is a special instance of the above Tr-GoF. HC rejects H_0 if a test statistic, denoted by HC_n^+, exceeds $\sqrt{2(1+\delta)\log\log n}$ for a small value $\delta>0$. This test statistic is related to our $S_n^+(2)$ as follows:

$$n\cdot S_n^+(2) = \sup_{r\in[p^+,1)} \frac{n}{2}\frac{(\mathbb{F}_n(r)-r)^2}{r(1-r)}\mathbf{1}_{\mathbb{F}_n(r)\ge r} = \frac{1}{2}(\mathrm{HC}_n^+)^2. \tag{9}$$

Our rejection rule aligns perfectly with HC as described in the literature (Cai et al., 2011; Cai & Wu, 2014; Donoho & Jin, 2004). See online supplementary material A.3 for further introduction and its theoretical property. Since HC using HC_n^+ can be viewed as a special case of Tr-GoF involving $nS_n^+(2)$, our primary analysis focuses on $nS_n^+(s)$ to cover a wide range of s. For completeness, we perform a simulation study on HC in online supplementary material B.2 and B.3.

4 Theoretical guarantees

In this section, we present the theoretical properties of Tr-GoF. We begin with several distributional assumptions on the LLM watermarks and human edits.

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Assumption 1 Let $\boldsymbol{P}_t$ denote the NTP distribution for w_t. Let $\mathcal{F}_{t-1} = \sigma(\{w_j, \zeta_j, \boldsymbol{P}_{j+1}\}_{j=1}^{t-1})$ be the σ-field generated by all text-generating randomness within the subtext $w_{1:(t-1)}$. We assume

(a) Perfect pseudorandomness: $\zeta_{1:n}$ are i.i.d. and $\zeta_t \perp \mathcal{F}_{t-1}$.
(b) Mixture of latent sources: $w_t = \mathcal{S}(\boldsymbol{P}_t, \zeta_t)$ with probability ε_n; otherwise, w_t is drawn independently from $\boldsymbol{P}_t$, and is independent of ζ_t when conditional on $\mathcal{F}_{t-1}$.

We briefly comment on Assumption 1. The σ-field $\mathcal{F}_{t-1}$ captures all related information from the subtext $w_{1:(t-1)}$, including the observed tokens, computed pseudorandom numbers, and their corresponding NTP distributions. Condition (a) states that the pseudorandom ζ_t, which either generates or verifies the watermark signal in w_t, behaves as an independent draw from a fixed distribution and is statistically independent of the past. This treatment—viewing pseudorandom as truly random—is standard in the watermarking literature, whether explicitly stated (X. Li et al., 2025; Piet et al., 2023; Wu et al., 2023; Zhao et al., 2025) or assumed implicitly (Fernandez et al., 2023; Kirchenbauer et al., 2023, 2024). Notably, Condition (a) aligns with the spirit of the Working Hypothesis proposed in our companion paper (X. Li et al., 2025). Condition (b) characterizes the relationship between the observed w_t and the computed ζ_t. The rationale behind it is that, since a verifier doesn't know which tokens have been edited, modelling w_t as a mixture of a watermarked token and a human-written one offers a principled way to capture this uncertainty. Specifically , w_t is either watermarked using ζ_t with probability ε_n, or it does not depend on ζ_t due to a broken watermark signal. This formulation enables theoretical analysis without modelling human behaviours, while still preserving the essential statistical structure for effective watermark detection.

4.1 Detectability

We begin by focussing on the challenging scenario where the watermark signal diminishes asymptotically. Specifically, with Assumption 1 and the following Assumption 2, we assume that both the non-null fraction, $\varepsilon_n \asymp n^{-p}$, and the distribution singularity, $\Delta_n \asymp n^{-q}$, decrease with the text length n. This setup draws inspiration from the classic framework of sparse detection problems (Donoho & Jin, 2004, 2015), where the signal strength of non-null effects also declines at a polynomial rate in sparse Gaussian mixture detection. Additionally, Assumption 2 can be relaxed to a weaker condition, which we discuss in Remark 6.

Assumption 2 Define $\Delta(\boldsymbol{P}) := 1 - \max_{w \in \mathcal{W}} P_w$ as the distribution singularity of $\boldsymbol{P}$. For any positive integer $t \le n$, we assume $\Delta(\boldsymbol{P}_t) = \Delta_n \asymp n^{-q}$ with $q \in [0, 1]$.

Remark 5 The distribution singularity $\Delta(\boldsymbol{P})$ is equivalent to the token entropy up to constant factors, the latter defined as $\mathrm{Ent}(\boldsymbol{P}) := \sum_{w \in \mathcal{W}} P_w \log \frac{1}{P_w}$. They are related via $\mathrm{Ent}(\boldsymbol{P}) = \Theta(\Delta(\boldsymbol{P}) \log \frac{1}{\Delta(\boldsymbol{P})})$. The proof is provided in Proposition A.2 in the online supplementary material.

The first question to address is whether there exists a statistical algorithm that can reliably solve the problem (4). As an extreme example, if $\varepsilon_n = 0$ or $\Delta_n = 0$, we would have $H_1^{\mathrm{mix}} = H_0$ and thus no test could differentiate $H_1^{(\mathrm{mix})}$ from H_0. In general, the detectability depends on the difference between the joint distributions of $Y_{1:n}$ under H_0 and $H_1^{(\mathrm{mix})}$. In the terminology of Donoho and Jin (2004), we say that H_0 and $H_1^{(\mathrm{mix})}$ merge asymptotically if the total variation distance between the joint distribution of $Y_{1:n}$ under H_0 and that under $H_1^{(\mathrm{mix})}$ tends to zero as the sequence length n goes to infinity. Otherwise, we say they separate asymptotically. The detection problem is detectable if and only if H_0 and $H_1^{(\mathrm{mix})}$ separate asymptotically.

The detectability behaves very differently in two regimes: the heavy edit regime where $\frac{1}{2} < p \le 1$ and the light edit regime where $0 < p \le \frac{1}{2}$. We begin by considering the heavy edit regime where $p \in (1/2, 1]$. Theorem 1 proves that no statistical test based on the observed pivotal statistics $Y_{1:n}$ can reliably detect the embedded watermark, if there are too many human edits. Notably, this impossibility result holds irrespective of the underlying NTP distribution $\boldsymbol{P}_t$'s. Therefore, in the heavy edit regime, watermark detection is impossible, regardless of the NTP distribution $\boldsymbol{P}_t$'s.

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Theorem 1 (Heavy edits case). Under Assumption 1, if $\frac{1}{2} < p \leq 1$, H_0 and $H_1^{(\text{mix})}$ merge asymptotically. For any test, the sum of Type I and Type II errors is 1 as $n \to \infty$.

The situation is much more involved when the edit is light where $p \in (0, 1/2]$. It turns out that the detectability of the embedded watermark further depends on the NTP distribution $\boldsymbol{P}_t$'s. To account for this, we introduce Assumption 2 which assumes the largest probability in all NTP distributions share the same value $1 - \Delta_n$ and $\Delta_n \asymp n^{-q}$.

Theorem 2 (Light edits case). Under Assumptions 1 and 2, let $0 \leq p \leq \frac{1}{2}$ and $0 \leq q \leq 1$.

(a) If $q + 2p > 1$, H_0 and $H_1^{(\text{mix})}$ merge asymptotically. Hence, for any test, the sum of Type I and Type II errors tends to 1 as $n \to \infty$.

(b) If $q + 2p < 1$, H_0 and $H_1^{(\text{mix})}$ separate asymptotically. For the likelihood-ratio test that rejects H_0 if the log-likelihood ratio is positive, the sum of Type I and Type II errors tends to 0 as $n \to \infty$.

Remark 6 (A weaker assumption for Theorem 2). For simplicity of presentation, we adopt Assumption 2, though the proof relies on a weaker condition: we assume that all NTP distributions $\boldsymbol{P}_t$ are either uniformly $\boldsymbol{P}_{1:n} \subset \mathcal{P}_{\Delta_n}$ or $\boldsymbol{P}_{1:n} \subset \mathcal{P}_{\Delta_n}^c$ according to (3). In the online supplementary material, we show that Theorem 2 holds under Assumption 1 and this weaker condition. Specifically, we prove that under Assumption 1, if $q + 2p > 1$ and $\boldsymbol{P}_{1:n} \subset \mathcal{P}_{\Delta_n}$, H_0 and $H_1^{(\text{mix})}$ merge asymptotically; conversely, if $q + 2p < 1$ and $\boldsymbol{P}_{1:n} \subset \mathcal{P}_{\Delta_n}^c$, H_0 and $H_1^{(\text{mix})}$ separate asymptotically. The same reasoning applies to the subsequent Theorem 3 and 4.

Theorem 2 shows that given $q + 2p < 1$, it is possible to distinguish between H_0 and $H_1^{(\text{mix})}$ asymptotically, with the likelihood-ratio test an effective detection method. Conversely, if $q + 2p > 1$, H_0 and $H_1^{(\text{mix})}$ become indistinguishable asymptotically. Hence, $q + 2p = 1$ represents the theoretical boundary distinguishing detectable from undetectable regions. Unfortunately, the likelihood-ratio test is impractical cause it relies on the unknown token distribution $\boldsymbol{P}_t$ and non-null fraction ε_n.

4.2 Adaptive optimality

An ideal optimal method should achieve the detection boundary $q + 2p = 1$ automatically in the previous difficult case, without requiring any problem-dependent knowledge such as p, q, or $\boldsymbol{P}_t$'s. This property is known as 'adaptive optimality' in the literature (Donoho & Jin, 2004, 2015), as it consistently works regardless of the underlying parameters and does not require this information. In the following theorem, we show that Tr-GoF achieves adaptive optimality.

Theorem 3 (Optimal adaptivity). Under Assumptions 1 and 2, if $q + 2p < 1$, for Tr-GoF with any $0 \leq c_n^+ \leq 1/n$ and $s \in [-1, 2]$, the sum of Type I and Type II errors tends to 0 as $n \to \infty$.

Due to the prevalence of sum-based detection rules in the literature (Aaronson, 2023; Fernandez et al., 2023; X. Li et al., 2025), we next examine whether these tests can achieve the same optimal adaptivity. We say a score function h is parameter-free if, for any y, $h(y)$ does not depend on Δ and ε.

Theorem 4 (Suboptimality of sum-based detection rules). Let Assumptions 1 and 2 hold. Consider the detection rule specified by h: $T_h(Y_{1:n}) = 1$ if $\sum_{t=1}^n h(Y_t) \geq n \cdot \mathbb{E}_0 h(Y) + \Theta(1) \cdot n^{\frac{1}{2}} a_n$, otherwise it equals 0, where $a_n \to \infty$ and $\frac{a_n}{n^\gamma} \to 0$ for any $\gamma > 0$.[3] For any score function that is (i) non-decreasing,

[3] The choice of a_n ensures that T_h has a vanishing Type I error. Examples include any polynomial function of $\log n$ or $\log \log n$ with positive coefficients.

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(ii) non-constant, (iii) parameter-free, and (iv) does not have discontinuities at both 0 and 1, the following results hold for T_h:

(a) If $q + p < \frac{1}{2}$, the sum of Type I and Type II errors tends to 0.
(b) If $q + p > \frac{1}{2}$, the sum of Type I and Type II errors tends to 1.

As shown in Theorem 4, for nearly any non-decreasing and parameter-free score function h, the sum-based detection rule it introduces is strictly suboptimal. Specifically, Theorem 4 shows that the detection boundary for these rules is $q + p = \frac{1}{2}$, rather than the optimal $q + 2p = 1$. Consequently, all existing sum-based detection rules fail to achieve the optimal detection boundary.

Corollary 4.1 Under Assumptions 1 and 2, the detection boundary for the existing score function $h \in \{h_{\text{ars}}, h_{\log}, h_{\text{ind},\delta}, h_{\text{opt},\Delta_0}\}$ with both $\delta, \Delta_0 \in (0, 1)$ is $q + p = \frac{1}{2}$.

4.3 Optimal efficiency rate

We now turn to the constant edit region, where $\Delta_n = \Delta \in (0, 1)$ and $\varepsilon_n \equiv \varepsilon \in (0, 1]$. In this scenario, $p = q = 0$ according to the notation $\Delta_n \asymp n^{-p}$ and $\varepsilon_n \asymp n^{-q}$. Theorem 1 implies that detection is always possible, as H_0 and $H_1^{(\text{mix})}$ asymptotically separate. Furthermore, Theorems 3 and 4 suggest that both Tr-GoF and sum-based detection rules are viable detection methods. Unlike the detection boundary, which assesses adaptivity to varying problem difficulties, we employ a different criterion to evaluate feasible detection methods.

X. Li et al. (2025) introduces a notion of test efficiency for watermark detection. The key idea is to define efficiency as the rate of exponential decrease in Type II errors for a fixed significance level α, considering the least-favourable NTP distribution within a belief class $\mathcal{P}$. The formal definition is provided in Definition 1.

Definition 1 ($\mathcal{P}$-efficiency X. Li et al., 2025). Consider the detection rule that rejects H_0 if $S(Y_{1:n})$ is larger than a critical value. Let $\gamma_{n,\alpha}$ be the critical value that ensures a Type I error of α for the problem (4), that is, $\mathbb{P}_0(S(Y_{1:n}) \geq \gamma_{n,\alpha}) = \alpha$ for all $n \geq 1$. For a given belief class $\mathcal{P}$, we define the following limit as the $\mathcal{P}$-efficiency rate of S and denote it by $R_{\mathcal{P}}(S)$:

$$\lim_{n\to\infty} \sup_{\boldsymbol{P}_t \in \mathcal{P}} \frac{1}{n} \log \mathbb{P}_1(S_n \leq \gamma_{n,\alpha}) = -R_{\mathcal{P}}(S).$$

This efficiency notion is referred to as class-dependent efficiency, as it depends on a given class $\mathcal{P}$, which characterizes the prior belief about the underlying NTP distributions $\boldsymbol{P}_t$'s. In their study, they set the prior class as the Δ-regular set $\mathcal{P}_\Delta$ (defined in (3)) and identify the optimal score function $h_{\text{opt},\Delta}$ for sum-based detection rules. It is natural to compute the $\mathcal{P}_\Delta$-efficiency rate for Tr-GoF, which provides a quantitative measure of their detection efficiency in the constant parameter region. We present the results in the following theorem.

Theorem 5 (Optimal $\mathcal{P}_\Delta$-efficiency). Assume Assumption 1 hold. Let $s \in (0, 1)$, $c_n^+ = 0$, $\varepsilon_n \equiv \varepsilon \in (0, 1]$ and $\Delta_n \equiv \Delta \in (0, 1)$. Given a dataset of pivotal statistics $Y_{1:n}$, it follows that

$$\sup_{\text{measurable } S} R_{\mathcal{P}_\Delta}(S) = D_{\text{KL}}(\mu_0, (1-\varepsilon)\mu_0 + \varepsilon\mu_{1,\boldsymbol{P}_\Delta^\star}) = R_{\mathcal{P}_\Delta}(S_n^+(s))$$

where $\boldsymbol{P}_\Delta^\star$ is the least-favourable NTP distribution defined by

$$\boldsymbol{P}_\Delta^\star = \left(\underbrace{1-\Delta, \ldots, 1-\Delta}_{\lfloor \frac{1}{1-\Delta} \rfloor \text{times}}, 1 - (1-\Delta) \cdot \left\lfloor \frac{1}{1-\Delta} \right\rfloor, 0, \ldots \right). \tag{10}$$

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Theorem 5 establishes both upper and lower bounds in terms of $\mathcal{P}_\Delta$-efficiency for the robust detection problem (4) under the constant parameter region. On one hand, it implies that the $\mathcal{P}_\Delta$-efficiency rate of any measurable function is upper-bounded by $D_{\mathrm{KL}}(\mu_0, (1-\varepsilon)\mu_0 + \varepsilon\mu_{1,\boldsymbol{P}_\Delta^\star})$. On the other hand, it shows that Tr-GoF achieves this optimal $\mathcal{P}_\Delta$-efficiency rate without any prior knowledge of ε and Δ. When $\varepsilon = 1$, this optimality is also achieved by the sum-based detection rule introduced by $h_{\mathrm{opt},\Delta}$ in X. Li et al. (2025). However, as can easily be seen, computing $h_{\mathrm{opt},\Delta}$ requires knowledge or correct belief about the value of Δ, which limits its practical applicability. Furthermore, when $\varepsilon < 1$, $h_{\mathrm{opt},\Delta}$ is no longer optimal because it does not consider the factor ε.

To illustrate this optimal efficiency further, Figure 4 presents the $\mathcal{P}_\Delta$-efficiency rates of different detection methods for different values of Δ. The left panel shows the case $\varepsilon = 1$ and the right panel shows $\varepsilon = 0.5$. Here, h_{ars}, $h_{\log}$, $h_{\mathrm{ind},\delta}$, and $h_{\mathrm{opt},\Delta_0}$ represent commonly used sum-based detection rules, with $\delta = 0.5$ and $\Delta_0 = 0.1$ set for illustration purposes. In both cases, Tr-GoF consistently achieves the optimal $\mathcal{P}_\Delta$-efficiency rate for all $\Delta \in [0.001, 1]$. In contrast, sum-based detection rules generally fail to reach the optimal efficiency rate across most values of Δ due to their lack of adaptivity. For example, when $\varepsilon = 1$, $h_{\mathrm{opt},\Delta_0}$ attains the optimal $\mathcal{P}_\Delta$-efficiency rate if $\Delta = \Delta_0$. However, once $\varepsilon < 1$, $h_{\mathrm{opt},\Delta_0}$ loses this optimality, as it no longer adapts to the reduced ε.

5 Simulations

In this section, our simulation studies first visualize the null and alternative distributions of $S_n^+(s)$ and then verify the empirical transition boundaries for different detection methods.

5.1 Experimental setup

We use a vocabulary of size $|\mathcal{W}|$ and model the pseudorandom variables $\zeta_{t,w}$ as true i.i.d. samples from $U(0, 1)$. We will explore the practical setting where ζ_t is computed via a hash function later. For given p, $q \in (0, 1]$, we set $\varepsilon_n = n^{-p}$ and $\Delta_n = n^{-q}$ where n is the text length. We use the following procedure to obtain samples of $\log(nS_n^+(s))$ under different settings:

(a) Draw n i.i.d. samples from $U(0, 1)$ to represent $Y_{1:n}$ in H_0 and calculate $\log(nS_n^+(s))$.
(b) Replace $\lceil n\varepsilon_n \rceil$ of the previous samples by the same number of samples from $F_{1,\boldsymbol{P}_t}$ where the NTP distributions $\boldsymbol{P}_t$ are i.i.d. generated. Then, calculate $\log(nS_n^+(s))$.
(c) Repeat Steps 1 and 2 N times. Obtain N independent samples of $\log(nS_n^+(s))$ under H_0 and H_1^{mix} respectively and create histograms of the simulated statistics.

In all subsequent experiments, we fix $N = 10^3$ while varying the values of $|\mathcal{W}|$, n, and the pair (p, q), which will be specified accordingly. We explore two methods for generating $\boldsymbol{P}_t$, denoted by M1 and M2. In particular, M1 randomly generates $\boldsymbol{P}_t$ by first setting its largest probability to $1 - \Delta_n$, and then configuring the remaining probabilities in $\boldsymbol{P}_t$ to adhere to Zipf's law (Zipf, 2016). In particular, we first i.i.d. sample $a_t \sim U(0.95, 1.5)$ and $b_t \sim U(0.01, 0.1)$, and then define $P_{t,w} = \Delta_n \cdot (w - 1 + b_t)^{-a_t}/C$, where $C = \sum_{w=2}^{|\mathcal{W}|} (w - 1 + b_t)^{-a_t}$ serves as the normalizing constant. M2 straightforwardly sets $\boldsymbol{P}_t$ to be equivalent to $(1 - \Delta_n, \frac{\Delta_n}{|\mathcal{W}|-1}, \ldots, \frac{\Delta_n}{|\mathcal{W}|-1})$.

5.2 Histograms of Tr-GoF statistics

To begin with, we investigate the empirical distribution of $\log(nS_n^+(s))$ under H_0 and H_1^{mix}. We set $|\mathcal{W}| = n = 10^3$ in this investigation and collect $N = 10^3$ samples of $\log(nS_n^+(s))$ for different settings. In Figure 5, we visualize the empirical density histograms of $\log(nS_n^+(s))$ for five values of $s \in \{2, 1.5, 1, 0.5, 0\}$ and three settings. We shade the rejection regions and mark the corresponding statistical powers as $1 - \beta$. We observe that

- For $0.5 \le s \le 1.5$, the null distributions of $\log(nS_n^+(s))$ exhibit similar shapes and supports. However, for $s \in \{0, 2\}$, there are noticeable changes in shape and support. Remarkably, the maximum value that $\log(nS_n^+(s))$ could take increases dramatically from 2 (at $s = 1$) to 4 or even 5 (when $s = 0$ or 2).
- For the majority of the s values, the empirical distribution of $\log(nS_n^+(s))$ under H_1^{mix} resembles its counterpart under H_0, albeit with a notable shift to the right. This shift can range from

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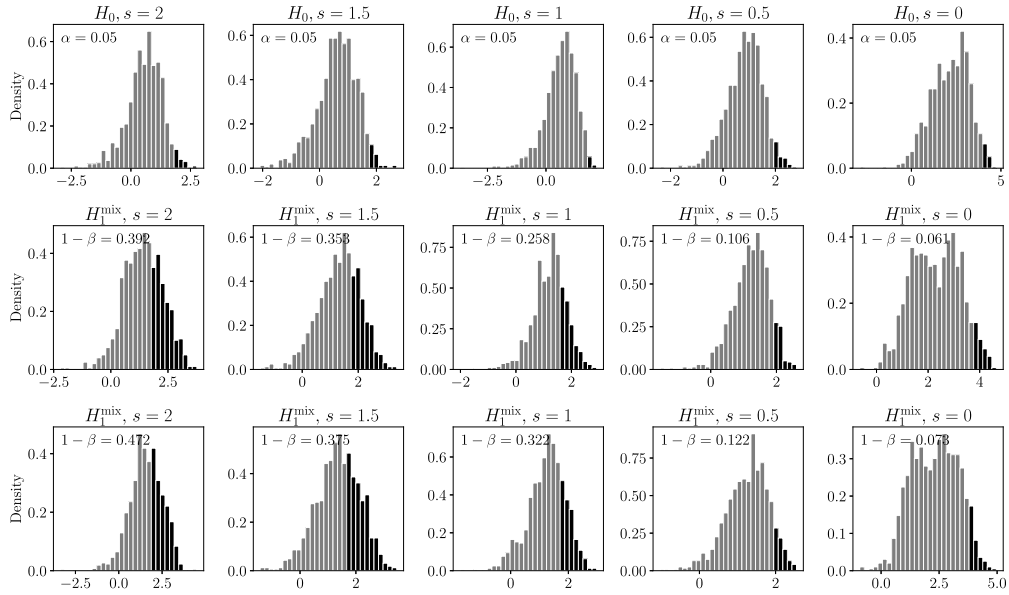


Figure 5. Density histograms and powers of $\log(nS_n^+(s))$ for different values of s with $c_n^+ = \frac{1}{n^2}$ and $(p, q) = (0.2, 0.5)$. The first row presents results under H_0, while the second and third rows correspond to H_1^{mix} in the M1 and M2 settings.

moderate (for $s \in \{0, 0.5\}$) to quite significant (for $s \in \{1, 1.5, 2\}$). It is this shift that enables the statistics $nS_n^+(s)$ to detect the distributional differences caused by embedded watermarks.

Even if we set $c_n^+ = 0$, the observed patterns remain consistent, except that the support under H_1^{mix} would be considerably larger due to the heavy-tailed behaviour of $p_{(1)}$. See online supplementary material B.1 for additional histogram results of $c_n^+ = 0$. Negative values of $s \in [-1, 0)$ yielded results nearly identical to $s = 0$ in both histograms and statistical power, so we omit them.

5.3 Detection boundary and optimal adaptivity

We then study the detection boundary of Tr-GoF. They reject H_0 when $\log(nS_n^+(s)) \geq C_{\text{GoF}}$ for a predetermined critical value C_{GoF} and a given value $s \in \{2, 1.5, 1, 0.5, 0\}$. We use the same setup introduced in Section 5.1 but with $|\mathcal{W}| = 5$ and focus on the M2 construction for NTP distributions. We aim to check how the smallest sum of Type I and Type II errors changes as we vary the other parameters namely (p, q), and n. Following the approach used in Cai et al. (2011), we compute the smallest sum errors by tuning the critical value C_{GoF} from a predetermined set. Here, we use the set $\mathcal{R}(0, 30, 10^3)$ where $\mathcal{R}(a, b, K)$ consist of K equally spaced points starting from a to b over an interval defined by $(K-1)$ divisions, that is, $\mathcal{R}(a, b, K) = \{a + \frac{k}{K-1} \cdot (b-a) : k = 0, 1, \ldots, K-1\}$.

Phase transition for a fixed p or q. We first fix either $q = 0.4$ or $p = 0.3$ and select a sample size n from $\{10^2, 10^3, 10^4\}$. According to Theorem 2, the detection boundary is given by $q + 2p = 1$, which suggests a transition at either $p = 0.3$ or $q = 0.4$. This prediction is validated by the first and second rows of Figure 6. For instance, when q is fixed at 0.4, the error sum $\alpha + \beta$ initially increases from zero and stabilizes at one as we increase p from zero to one. The transition point occurs around $p = 0.3$ and aligns well with the red dashed line. Furthermore, larger sample sizes make the alignment with the theoretical prediction more pronounced.

Adaptive optimality of Tr-GoF. To accurately capture the empirical detection boundary, we use $n = 10^4$ independent samples to calculate $\log(nS_n^+(s))$. For any $p \in \mathcal{R}(0.01, 1, 20)$ and

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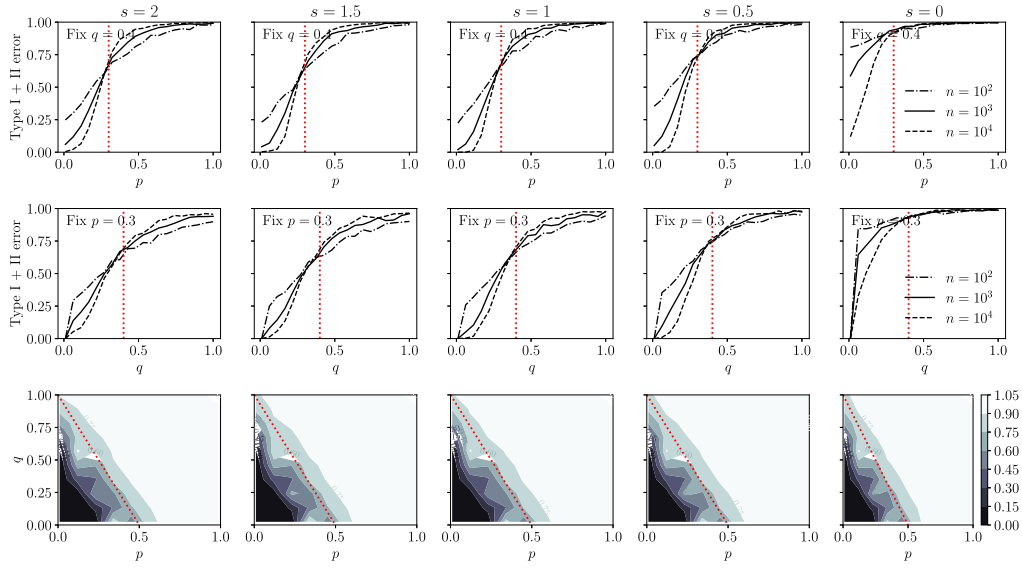


Figure 6. Empirical detection boundaries of Tr-GoF for different values of s. Each column corresponds to results for a specific value of s. The first two rows display the sum of Type I and II errors, with q fixed at 0.4 and p at 0.3, respectively. The last row features the contour plot which illustrates the sum of errors across the domain $[0, 1]^2$, with $n = 10^4$. Red dotted lines present the theoretical detection boundaries. All the results are averaged over $N = 10^3$ independent trials.

$q \in \mathcal{R}(\log_n \frac{|\mathcal{W}|}{|\mathcal{W}|-1}, 1, 20)$,[4] we compute the smallest sum of Type I and Type II errors by tuning the critical value as mentioned earlier. These results are displayed in the bottom row of Figure 6. Here, darker areas indicate lower error sums, while lighter regions present higher error sums. A red dashed line represents the theoretical detection boundary $q + 2p = 1$. Most darker regions are below this boundary, while lighter regions are above it. This empirical boundary aligns well with the theoretical prediction in Theorem 3, no matter what the value of s we choose.

Suboptimality of existing detection methods. For any function $h \in \{h_{\text{ars}}, h_{\log}, h_{\text{ind},\delta}, h_{\text{opt},\Delta_0}\}$, we consider the following sum-based detection rule specified by h:

$$T_h(Y_{1:n}) = \begin{cases} 1 & \text{if } \sum_{t=1}^n h(Y_t) \geq n \cdot \mathbb{E}_0 h(Y) + C_{\text{sum}} \cdot n^{\frac{1}{2}} \log n, \\ 0 & \text{if } \sum_{t=1}^n h(Y_t) < n \cdot \mathbb{E}_0 h(Y) + C_{\text{sum}} \cdot n^{\frac{1}{2}} \log n. \end{cases}$$

The expectation $\mathbb{E}_0 h(Y)$ for different h's has been listed in Table 1 in X. Li et al. (2025). Mirroring our approach with Tr-GoF, for each detection rule T_h, we tune the parameter C_{sum} from a given set to obtain the smallest sum of Type I and Type II errors. This set for h_{ars} is $\mathcal{R}(8, 60, 10^3)$, for $h_{\log}$ is $\mathcal{R}(-20, 0, 10^3)$, and for $h_{\text{ind},0.5}$, $h_{\text{opt},0.1}$ is $\mathcal{R}(-10, 10, 10^3)$. The findings are presented in Figure 7, which was created using the same procedure as the contour plot in Figure 6. Observations reveal that the empirical boundaries closely match the theoretical prediction of $p + q = \frac{1}{2}$, represented by yellow dotted lines. This alignment corroborates our Theorem 4. In contrast, the red dotted line represents the optimal detection boundary $q + 2p = 1$. The discrepancy between the yellow and red dotted lines shows the suboptimality of sum-based tests.

6 Experiments on open-source LLMs

6.1 Experiment setup

We follow the experimental setup from Kuditipudi et al. (2024) and X. Li et al. (2025). We begin by sampling 1,000 documents from the news-like C4 dataset (Raffel et al., 2020), which serve as

[4] $\log_n \frac{|\mathcal{W}|}{|\mathcal{W}|-1} \leq q$ is solved from the fact $1 - n^{-q} \geq \frac{1}{|\mathcal{W}|}$ which always holds due to the pigeonhole principle.

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Table 1. The edit tolerance limits (%) for detection methods on the OPT-1.3B model. The best results are highlighted in bold.

Task	Edit types	$s=1$	$s=1.5$	$s=2$	$h_{\log}$	h_{ars}	$h_{\text{opt},0.3}$	$h_{\text{opt},0.2}$	$h_{\text{opt},0.1}$
Poem Recitation	Substitution	37.16	38.8	**39.49**	25.04	37.96	27.83	30.04	33.87
	Insertion	42.15	45.25	**45.64**	26.46	44.12	30.38	33.32	37.78
	Deletion	39.78	41.85	**42.67**	23.21	41.92	27.08	30.12	33.76
Poem Generation	Substitution	36.89	38.72	**38.9**	24.71	38.64	27.47	30.04	33.49
	Insertion	40.08	42.52	**43.09**	25.54	41.65	29.49	32.75	36.28
	Deletion	39.79	41.83	**42.42**	26.38	40.99	29.59	32.3	35.15

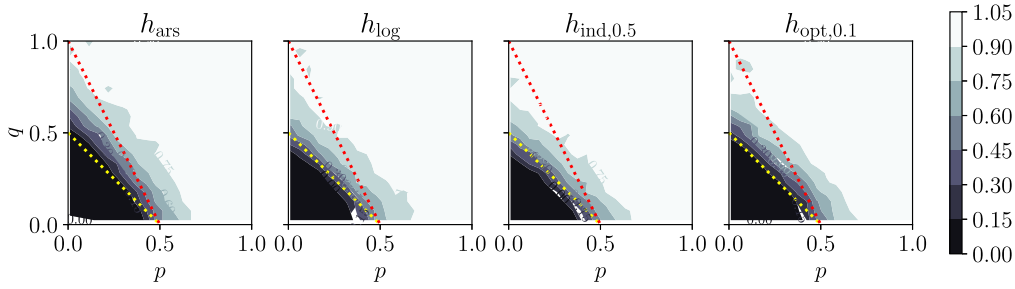


Figure 7. Empirical detection boundaries for sum-based detection rules T_h's.

the initial prompts. Using these prompts, we then ask two language models, the OPT-1.3B model (Zhang et al., 2022) and Sheared-LLaMA-2.7B (Xia et al., 2024), to generate an additional $n=400$ tokens for each document. Our evaluation focuses on two key aspects:

(a) **Statistical power:** To evaluate the statistical power of different detection methods, we set the significance level at $\alpha=0.01$ by tuning the critical values, either using central limit theorem predictions or Monte Carlo simulations. We assess the actual Type I errors using unwatermarked texts and Type II errors using watermarked texts.

(b) **Robustness to edits:** Almost any editing method tends to weaken watermark signals. We focus on three representative types of edits (Kuditipudi et al., 2024): random edits (including substitution, insertion, and deletion), adversarial edits, and roundtrip translation. In random edits, a random fraction of the watermarked tokens is replaced, inserted, or deleted, with substitutions and insertions involving tokens uniformly selected from the vocabulary $\mathcal{W}$. Adversarial edits, by contrast, are more targeted, selectively modifying watermarked tokens to maximize the removal of watermark signals within a given edit budget. Finally, roundtrip translation involves translating the text from English to French and back to English using another language model. Random and adversarial edits enable systematic control over the level of edits, while roundtrip translation is more likely to be encountered in practise.

We compare the Tr-GoF test across three different values of s from the set $\{1, 1.5, 2\}$ alongside three sum-based detection rules, each specified by a particular score function h. These include: (i) Aaronson's function $h_{\text{ars}}(y)=-\log(1-y)$ (Aaronson, 2023), (ii) the logarithmic function $h_{\log}(y)=\log y$ (Fernandez et al., 2023; Kuditipudi et al., 2024), and (iii) the optimal least-favourable function (X. Li et al., 2025) with Δ a user-specified regularity parameter Among $\Delta\in\{0.1, 0.2, 0.3\}$, $\Delta=0.1$ performs best and is used in all subsequent experiments. In all experiments, we use 1-sequence repeated context masking (Dathathri et al., 2024). This approach

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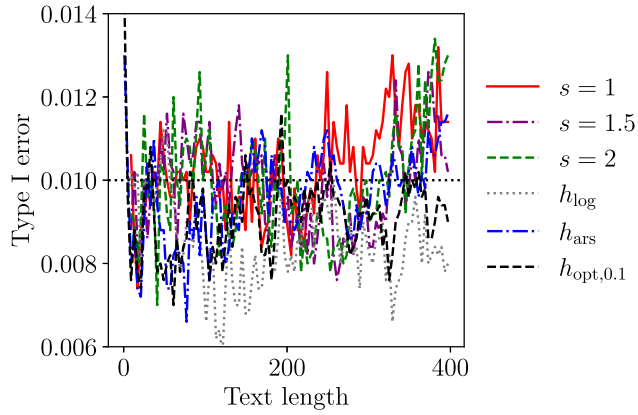


Figure 8. Empirical Type I errors.

watermarks a token only when the current text window is unique within the generation history, aiming to preserve text quality. In the following, we present our numerical results on the OPT-1.3B model for illustrative purposes, as the results for Sheared-LLaMA-2.7B are similar and are provided in the supplementary. Additional experimental details are also deferred to the online supplementary material.

6.2 Statistical power

We first evaluate the statistical power of considered detection methods using unmodified texts. Unlike in simulation studies, we cannot manipulate each NTP distribution $\boldsymbol{P}_t$ in language model experiments to ensure they are Δ-regular. However, Δ-regularity correlates closely with the temperature parameter in LLMs (Ackley et al., 1985). The temperature parameter modulates the raw outputs of the model's final layer (also known as the logit layer) before the softmax function is applied. Typically, a high temperature yields more uniform probabilities, encouraging diverse generations, while a low temperature sharpens the distribution, emphasizing the most probable prediction and thus favouring the greedy generation. As an approximation, we use four temperatures {0.1, 0.3, 0.5, 0.7} to check the effect of Δ-regularity on statistical power. The evaluation results are in Figures 8 and 9. Results on the Sheared-LLaMA-2.7B model are similar and documented in online supplementary material C.2.

Type I error control. We begin by examining Type I error control, using 5000 unwatermarked texts sampled from the C4 dataset as human-written data and evaluating Type I error at the significance level $\alpha = 0.01$. Figure 8 shows how the empirical Type I error varies with increasing text lengths, revealing a consistent pattern: across all detection methods, empirical Type I errors remain well-controlled within the interval [0.006, 0.014]. This suggests that the pseudorandom variables effectively mimic the behaviour of true random variables, enabling their empirical performance to closely align with theoretical expectations.

Type II error decay. We then examine the decay of Type II errors at a significance level of $\alpha = 0.01$, shown in the top row of Figure 9. The bottom row displays a trade-off function illustrating detection performance across all critical values, extending beyond the fixed $\alpha = 0.01$ case. From left to right, the columns show increasing temperatures from 0.1 to 0.7. We observe:

(a) *Tr-GoF excels in the low-temperature region.* When the temperature is relatively low (that is, 0.1, 0.3), the Type II errors for the Tr-GoF test decrease more rapidly and eventually fall below those of all baseline detection rules (when $s \in \{1, 1.5\}$). For instance, at a temperature of 0.1 and using 400 watermarked tokens, the baseline detection methods obtain a Type II error of approximately 0.6, whereas the best Tr-GoF test exhibits Type II errors around 0.3.

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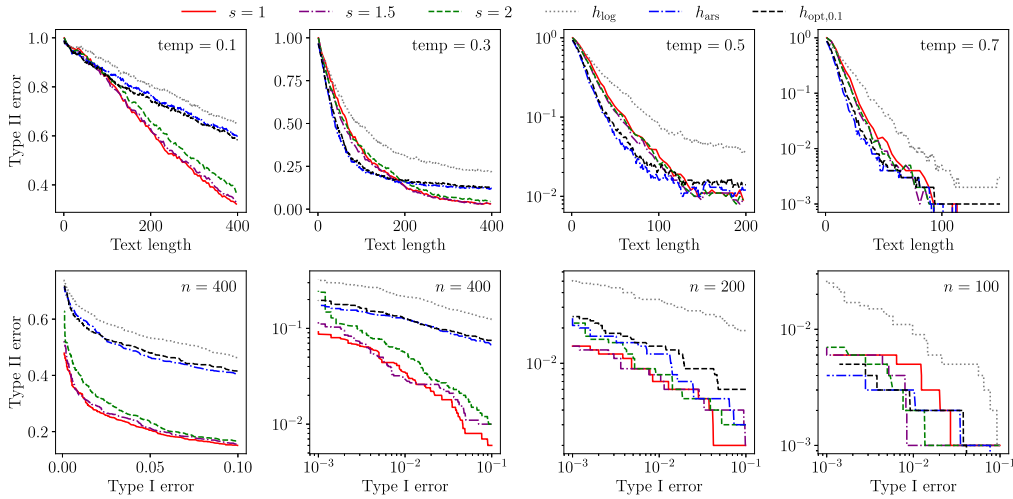


Figure 9. Empirical Type II errors (top row) across different detection rules applied to the Gumbel-max watermark. The bottom row illustrates the trade-off function in the log-log scale for a specific length n. The temperatures used, from left to right columns, are 0.1, 0.3, 0.5, and 0.7, respectively.

Interestingly, although the optimal least-favourable function $h_{\text{opt},0.1}$ performs (slightly) better than h_{ars}, it is still inferior to the Tr-GoF test. The superior performance of the Tr-GoF test is further evidenced by the trade-off functions, where the Tr-GoF test consistently outperforms other methods for nearly all given Type I errors. However, this does not contradict the claimed optimality of h_{opt,Δ_0} in X. Li et al. (2025). This is because the optimality is defined for the least-favourable case, but in practise, the conditions are not adversarial enough for it to matter.

(b) *Tr-GoF performs comparably in the high-temperature region.* At relatively high temperatures (that is, 0.5 and 0.7), the Tr-GoF test achieves Type II error decay comparable to the baseline h_{ars}. At a temperature of 0.5, most detection methods reach a Type II error of 0.01 with only 100 watermarked tokens. The trade-off function indicates that the Tr-GoF test maintains a slightly lower trade-off for small Type I errors (>0.001). At a temperature of 0.7, nearly all Type II errors decay to 0.01 with approximately 50 watermarked tokens. The improved detection effectiveness at higher temperatures stems from the increased variability in token generation, which strengthens statistical signals and facilitates watermark detection.

6.3 Robustness evaluation

Robustness to random edits. We examine three types of edits: random substitution, insertion, and deletion. For each specified edit method and a given fraction, we randomly modify the corresponding fraction of tokens in the watermarked text. We set the Type I error to $\alpha = 0.01$ and investigate how the Type II error changes with the edit fraction across four different temperatures: {0.1, 0.3, 0.7, 1}. This approach allows us to assess the robustness of watermark detection methods under various text edit conditions and temperature settings. Generally, with a text window of size m, modifying a single token can affect the computation of up to m pseudorandom numbers. Therefore, a window size of m can always withstand an edit fraction of up to $1/m$, as not all watermarked signals are removed. In our experiments, we set $m = 5$. The results of random edits are shown in Figure 10, with the top, middle, and bottom rows corresponding to random substitution, insertion, and deletion, respectively. Across all edit types, we observe a consistent pattern: any edit increases the Type II error for all detection methods. At lower temperatures, the error rate increases sharply with rising edit fractions, while at higher temperatures, the increase is more gradual, indicating reduced sensitivity to edits.

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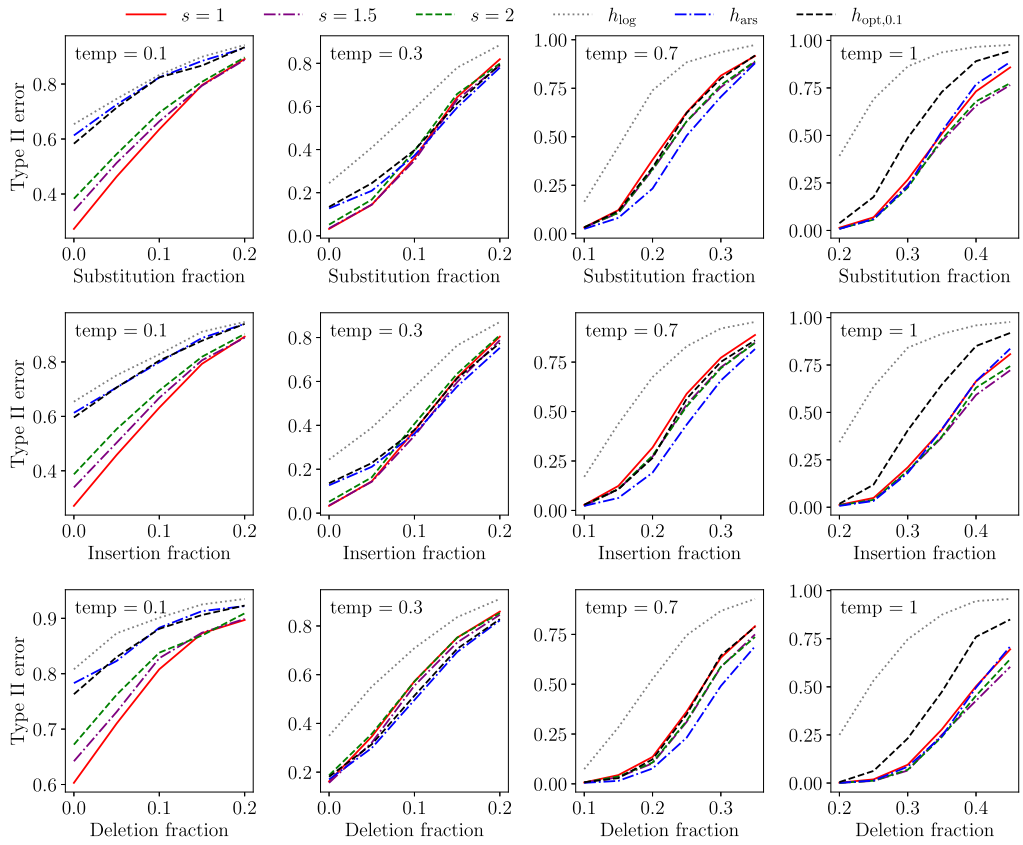


Figure 10. Effect of three random edits on Type II error across different temperatures at a fixed Type I error of $\alpha = 0.01$. The top, middle, and bottom plots correspond to random substitution, insertion, and deletion, respectively.

(a) *Tr-GoF demonstrates better robustness at lower temperatures.* The Tr-GoF test consistently achieves the lowest Type II error rates across all three edit types at lower temperatures (that is, 0.1, 0.3), outperforming other methods such as $h_{\text{opt},0.1}$ and h_{ars}. This makes it particularly effective when the LLM outputs are more deterministic.

(b) *Tr-GoF maintains comparable robustness at higher temperatures.* While the Tr-GoF test slightly trails behind h_{ars} at a temperature of 0.7, it remains competitive, with an average difference in Type II error of only 0.02 compared to h_{ars} across different edit fractions. At a temperature of 0.7, $s = 2$ slightly outperforms the others. This suggests that, even in high-temperature scenarios, the Tr-GoF test maintains robust performance and adapts effectively to the impact of temperature on detection difficulty.

Edit tolerance limit. We define the edit tolerance limit as the largest fraction of edits that can be applied to a watermarked text with the detection method still rejecting the null hypothesis H_0. In general, the higher the edit tolerance limit, the more robust the detection method, as it is more sensitive to the weak watermark signal.

We compute the edit tolerance limits for different detection methods across two tasks: (a) poem recitation, where the LLM is asked to recite an existing poem, and (b) poem generation, where the LLM generates a new poem in the style of a given one. The latter task is more open-ended, leading to more stochastic and regular generation. The results, averaged over 100 popular poems, are reported in Table 1, with the highest values (in percentage) highlighted in bold. Notably, the Tr-GoF test with $s = 2$ consistently achieves the highest edit tolerance across all three edit types and both

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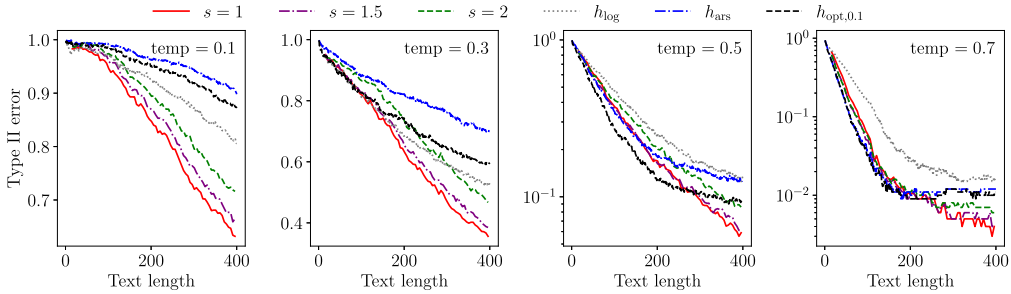


Figure 11. Effect of adversarial edits on Type II error across different temperatures.

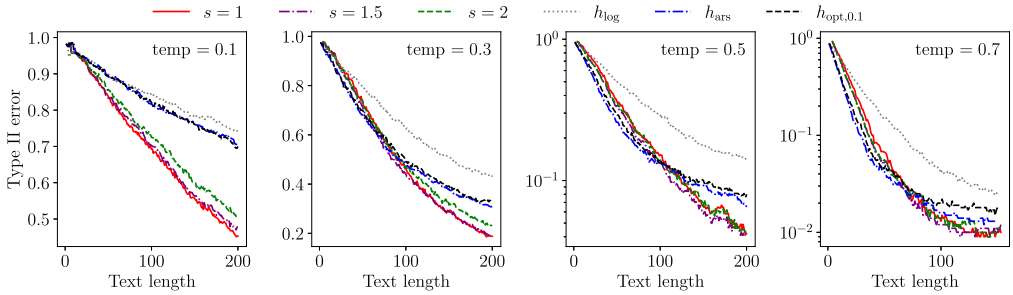


Figure 12. Effect of roundtrip translation on Type II error across different temperatures.

tasks. These findings align with the results shown in Figure 10. Similar results for the Sheared-LLaMA-2.7B model are provided in online supplementary material C.3.

Robustness to adversarial edits. In adversarial edits, we assume the human user knows the hash function $\mathcal{A}$ and the secret key `Key`, allowing them to selectively replace tokens with the strongest watermark signals to evade detection. To approximate this behaviour, we use the following procedure: for the LLM-generated response, the user first computes all corresponding pivotal statistics, identifies a given fraction of tokens with the highest pivotal statistics, and replaces them with randomly selected tokens. This targeted replacement is more disruptive than random edits. Results for a 5% replacement are shown in Figure 11, with results for other fractions in online supplementary material C.3.

In this adversarial setting, the Tr-GoF test demonstrates steady robustness, consistently achieving a lower Type II error across most temperature settings. Notably, h_{ars} is less resilient under adversarial edits, while $h_{\text{opt},0.1}$ performs better. This increased robustness in $h_{\text{opt},0.1}$ likely stems from its design through minimax optimization, which enhances its ability to withstand adversarial edits.

Robustness to roundtrip translation. In the roundtrip translation, we cannot control the edit level. By maintaining a fixed significance level of $\alpha = 0.01$, we examine how the Type II error varies with the modified text length and different temperatures. The results are present in Figure 12. Using $n = 200$ tokens, we observe that the final Type II errors decrease as the temperature increases, suggesting that higher temperatures facilitate easier detection. In the low-temperature range, the Tr-GoF test with $s = 1$ consistently outperforms all other detection methods. Conversely, in the high-temperature range, Tr-GoF tests with $s \in \{1, 1.5\}$ achieve comparable or occasionally superior performance to the previous h_{ars}. These experiments underscore the robust performance and detection efficiency of Tr-GoF for Gumbel-max watermarks.

7 Discussion

In this paper, we have developed an adaptive and robust method for detecting watermarks in text generated by LLMs but subsequently edited by humans, along with theoretical guarantees

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corroborated by numerical experiments. We start by proposing a procedure for modelling the human editing process, which prompts us to formulate this problem as sparse mixture detection. Our method, which we call the Tr-GoF test, is shown to achieve the optimal detection boundary in the regime of increasing edit level and vanishing regularity of NTP distributions under the framework of X. Li et al. (2025). In contrast, we show that sum-based detection rules are provably suboptimal in the sense that the detection region is strictly smaller than that of the Tr-GoF test. Additionally, we show that the Tr-GoF test continues to provide high detection efficiency when the edit level and regularity of NTP distributions remain constant. In contrast, sum-based detection rules fail to achieve robustness in either regime due to their inability to adapt to unknown specifics of problem instances.

Our findings open new avenues for the robust detection of LLM watermarks. First, although we focus on the Gumbel-max watermark, which is perhaps the most influential, it would be valuable to investigate robust detection for other watermarking schemes (Xie et al., 2025), such as the inverse transform watermark Kuditipudi et al. (2024) and the green-red list watermark Kirchenbauer et al. (2023). In these cases, the truncation technique underlying Tr-GoF could offer useful insights. More broadly, future research might consider alternative nonparametric testing approaches (D'Agostino, 2017; Nikitin, 1995), such as the Kolmogorov–Smirnov test, Pearson's Chi-square test, and the Khmaladze–Aki statistic (Khmaladze, 1982; Podkorytova, 1994), which may offer adaptivity comparable to or even surpassing goodness-of-fit tests in certain parameter regimes. Empirical evaluation of these classical nonparametric methods could motivate the development of new procedures tailored to specific challenges, such as high-temperature regimes, where Tr-GoF may undergo a decline in detection power. Additionally, detection power could be enhanced by incorporating spatial information from watermark signals, as human edits often form clusters within text. From a theoretical standpoint, it is noteworthy that the optimal detection boundary $q + 2p = 1$ is achieved through pivotal statistics, suggesting that investigating an information-theoretic detection boundary could yield interesting insights. Lastly, for interpretive purposes, estimating the watermark fraction may offer insights into the degree of human contribution in content generated collaboratively by LLMs and humans.

Conflicts of interest: The authors declare no conflicts of interest.

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Data availability

The data underlying this article are drawn from the news-like C4 dataset, which is publicly available through the Hugging Face API at https://huggingface.co/datasets/allenai/c4. The computer code and algorithms used in this study are openly accessible at https://github.com/lx10077/TrGoF.

Supplementary material

Supplementary material is available online at *Journal of the Royal Statistical Society: Series B*.

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